

# BioSpecInfo

## Functional Specifications — BioSpecInfo

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# Functional Specifications — BioSpecInfo

| Field             | Value                                                                              |
|-------------------|------------------------------------------------------------------------------------|
| Software          | BioSpecInfo                                                                        |
| Version described | bsi-v176                                                                           |
| Purpose           | Describe what the product does, for whom, under which rules and with which limits. |

## 1. Product description

BioSpecInfo is an **interactive cheminformatics** platform for university study of chemistry, biochemistry, pharmacology and astrochemistry.

The whole scientific logic — interpretation of molecular structures, spectral prediction, calculations, 2D and 3D visualisation — runs **in the student's browser**. There is no server: the application installs as an app, works with no network, and transmits no data.

### 1.1 Objectives

| Objective                     | How it is pursued                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Total accessibility           | No registration, no cost, no server. Whoever opens the link is already using it.                                                  |
| Independence from the network | Everything needed is bundled and cached: it works in a lecture hall, on a train, in a library with no signal.                     |
| Verifiable correctness        | Scientific data are put through automatic checks; the limits of the models are declared rather than hidden.                       |
| Privacy by construction       | There is no backend that could collect data: the absence of collection is not a promise, it is a consequence of the architecture. |

### 1.2 What the product is not

|                                       |                                                                          |
|---------------------------------------|--------------------------------------------------------------------------|
| Not a medical device                  | It does not support diagnosis, prescription or dosing (Reg. EU 2017/745) |
| Not a certified analytical instrument | The spectra are predictions, not measurements                            |
| Not a collaborative platform          | No accounts, no sharing, no synchronisation between devices              |

## 2. Actors

| Actor   | Description                                   | What they can do                                                                    |
|---------|-----------------------------------------------|-------------------------------------------------------------------------------------|
| Student | Primary user. Studying for a university exam. | Every function: consultation, calculation, quizzes, spaced repetition, AI assistant |

| Actor              | Description                               | What they can do                                                       |
|--------------------|-------------------------------------------|------------------------------------------------------------------------|
| Teacher            | Uses the app as teaching support          | The same functions; uploading their own material into the File Manager |
| Technical assessor | A company or committee examining the work | Reading the documentation and re-running the verification benches      |

There are no technical roles: there is no authentication, so there is no distinction of permissions. The File Manager has a local protection that is a **deterrent**, not an access control (see §6).

## 3. Functional areas

The application has **88 sections**, grouped by area of study.

### 3.1 Spectroscopy

| Function                   | Description                                                                                                                                                 |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Spectroscopy centre</b> | A molecule is entered (SMILES, name, formula) and one obtains IR, $^1\text{H}/^{13}\text{C}$ NMR, MS, UV-Vis, Raman, 2D and 3D structure, functional groups |
| <b>Visual IR spectra</b>   | Characteristic bands by class of compound                                                                                                                   |
| <b>Mass spectrum</b>       | Fragmentation patterns and diagnostic rules                                                                                                                 |
| <b>UV-Vis</b>              | Chromophores, $\lambda_{\text{max}}$ , Woodward's rules                                                                                                     |
| <b>NOE and 2D NMR</b>      | Spatial correlations                                                                                                                                        |
| <b>Reference tables</b>    | Tabulated values for quick consultation                                                                                                                     |
| <b>Spectrum quiz</b>       | Recognising the functional group from a spectrum                                                                                                            |

### 3.2 General and physical chemistry

Interactive periodic table · chemical calculator · reaction balancing · degrees of unsaturation · distribution diagrams · VSEPR · molecular orbitals · advanced thermodynamics · electrochemistry (Nernst) · pKa and buffers · laboratory calculator · constants and formulae.

### 3.2-bis Separation — chromatography

**Chromatography** section: a chromatogram computed from the parameters (theoretical plates, retention factor, selectivity, dead time) with an overview and an expanded view of the peak pair; a **van Deemter** curve with the three terms separated and the minimum computed in closed form; a **Kováts retention index** calculator; a table of the quantities with their working ranges; GC/HPLC comparison; a guide to detectors; diagnostics for peak-shape defects.

**Rule applied.** The chromatogram is not an illustration: the Gaussians have  $\sigma = t_{R}/\sqrt{N}$  and the resolution shown is the one that would be measured on those peaks. The section explicitly declares what it does **not** do — predict a molecule's retention time from its structure, which would require experimental parameters of the stationary phase that cannot be derived from the SMILES alone.

### 3.2-ter Calibration and uncertainty

**Calibration** section: ordinary least-squares regression on data pasted from a spreadsheet, with slope and intercept **accompanied by their standard deviation**; a plot of the line with a 95 % confidence band and a **residual plot**; the unknown's concentration with a full confidence interval (the three sources 1/M, 1/N and distance from the centroid are shown separately); **LOD and LOQ** per ICH Q2(R1); uncertainty propagation indicating which contribution dominates; significant-figure rules.

Two diagnostics that an  $r^2$  does not provide:

| Diagnostic           | Method                                                                              | Why $r^2$ is not enough                                                                                                                                |
|----------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Non-linearity</b> | F-test between line and parabola on the same data                                   | An $r^2$ of 0.984 accompanies an obvious saturation curve: $r^2$ measures how much variance the line explains, not whether the line is the right model |
| <b>Outlier</b>       | <b>Deleted</b> residual: the judging line is rebuilt without the point being judged | The naive criterion `                                                                                                                                  |

**The second row is a trap met while writing the section**, not a textbook case: the  $3\sigma$  criterion was already written, and the example meant to demonstrate it was not being flagged. The check passed without seeing anything.

### 3.2-quater Cheminformatics and QSAR

**Cheminformatics** section: a workbench that goes from the raw set to a verdict on the model, entirely in the browser. SMILES are pasted in — with a name and an activity, where there are any — and six panels return what a cheminformatics group produces in a day.

| Panel                        | What it does                                                                                                                                                                                                      |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Descriptors</b>           | Standardisation (largest fragment, canonical SMILES, deduplication), 43 RDKit descriptors, Lipinski's rule of five, Veber filter, QED recomputed — MinimalLib does not expose it                                  |
| <b>Similarity and groups</b> | Morgan/ECFP4, RDKit, pattern and MACCS fingerprints; Tanimoto and Dice; similarity matrix; nearest neighbours; <b>Butina clustering</b> and <b>MaxMin</b> selection for diversity                                 |
| <b>Chemical space</b>        | PCA on the standardised descriptors, with explained variance per component and the <b>loadings</b> that say which descriptor moves which axis; <b>Bemis-Murcko</b> scaffolds with the molecule count per scaffold |

| Panel             | What it does                                                                                                                                                                                                                                                                                                                                           |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QSAR model        | Kernel ridge (Tanimoto kernel over the fingerprints) or logistic regression; <b>scaffold split</b> alongside the random one; <b>cross-validation over 3, 5 or 10 scaffold-grouped folds</b> ; $R^2$ , RMSE, MAE, Pearson for regression, ROC-AUC and MCC for classification; <b>applicability domain</b> by distance to the nearest training neighbour |
| Activity cliffs   | Structurally close pairs with distant activities, ranked by <b>SALI</b> : these are the pairs every model gets wrong, and it is honest to show them                                                                                                                                                                                                    |
| Structural alerts | PAINS and Brenk, with the matched fragment highlighted and the reason it is flagged                                                                                                                                                                                                                                                                    |

Three choices that separate a workbench from a demonstration:

| Choice                              | Method                                                                                             | Why                                                                                                                                                                                                             |
|-------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scaffold split                      | The test set contains <b>only scaffolds never seen</b> in training                                 | A random split puts close analogues on both sides: the model recognises rather than predicts, and the $R^2$ that comes out does not survive the first new molecule                                              |
| Null model by scrambling            | The same model is retrained on <b>shuffled labels</b> , eight times; the verdict is the comparison | An $R^2$ of 0.4 can arise by chance when the molecules are few. If the random twins reach the same score, the model has learnt nothing — and the panel <b>says so</b> , instead of showing only the good number |
| Deduplication on the canonical form | <code>OC(=O)C</code> and <code>CC(=O)O</code> are the same molecule                                | The same molecule in training and in test is an information leak, and the written string is not enough to see it                                                                                                |

**The verdict is written out in words, not left to the reader.** The QSAR panel does not stop at the metrics: it states whether the model beat all its random twins, and by how much. Measured on the 28-inhibitor example:  $R^2$  0.861 against a maximum of 0.574 among the eight twins — margin 0.287. On the same molecules with random labels the model does **not** beat the twins, and the panel says so in the same way.

## The results can be taken away

Two exports, both formed in memory by the browser: there is no network request, which is why the note beside the buttons can say that no data leaves the device without asking anyone to take it on trust.

| Export              | Content                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CSV table           | One row per molecule: name, canonical and original SMILES, activity, scaffold, nine descriptors, QED, Lipinski violations, Veber outcome, alerts found. With cross-validation active, <b>predicted</b> and <b>residual</b> are added — and only then, because cross-validation is the one scheme that gives every molecule a prediction from a model that had not seen it |
| Method report (.md) | What is needed to <b>redo</b> the same analysis: RDKit version, fingerprint, split type, number of molecules and scaffolds, <b>the seed</b> , the metrics, the null-model verdict, the folds one by one, and the declared limits                                                                                                                                          |

**Why the seed is in the report.** Without it, "random split" and "scrambled labels" mean nothing: two runs would give different numbers and nobody could say which was the right one. The generator is a seeded mulberry32, and the same seed gives the same folds.

## A measurement no other bench could make

`verifica-accessibilita` walks the 88 sections and measures what is **visible**. The six panels of this section live inside a container that stays `display:none` until the analysis has been run: that bench never saw them, and its "contrast 0" was not speaking about them.

`test_cheminfo` runs the analysis for real, opens the six panels in turn and measures WCAG contrast on every piece of text that appears there — **460 elements, 0 defects**. The formulas are rewritten inside the bench rather than taken from the application: a bench that asked the code under test how good its own contrast was would be measuring nothing.

The engine lives in `bsi-cheminfo.js`, is exposed as `window.BSIChem` and is verified by the `test_cheminfo` bench (106 checks) against hand-computable values.

## 3.3 Organic chemistry

Retrosynthesis (98 exercises) · synthesis and mechanisms with electron arrows · animated mechanisms · structure editor · *Organic Chemistry 1+2* module with 25 modules.

## 3.4 Biochemistry

Amino acids · biomolecules · animated metabolic pathways · enzyme kinetics (Michaelis-Menten) · 3D macromolecules · interactive biological structures · biochemistry guide · exam biochemistry.

The **biochemistry guide** is a document of its own (`Biochimica_Guida_Definitiva.html`, 8.2 MB) loaded into a frame by the section: 12 chapters plus the index, 78 subsections, 98 embedded figures, and two chapters devoted to the exam — "Oral exam guide" and "Real exam questions and correct answers". It contains **no scripts and makes no network requests**: it cannot raise JavaScript errors, and the alternative text of the 98 figures comes from the captions the author wrote, not from invented descriptions (see `docs/06 §3-ter`).

## 3.5 Pharmacology

| Function         | Description                                                                                       |
|------------------|---------------------------------------------------------------------------------------------------|
| 3D Atlas         | One chooses a body region, a disease and a drug: the model highlights the target organ            |
| Database         | <b>178 drugs</b> with structure, molecular weight, mechanism, indications, adverse effects, class |
| Interactions     | Checking interactions between active ingredients                                                  |
| Pharmacokinetics | Calculator for half-life, clearance, volume of distribution                                       |
| Clinical cases   | Diagnosis-therapy paths for teaching purposes                                                     |

### 3.6 Astrochemistry

Molecules of the interstellar medium · exoplanets and JWST data · nebulae · stellar spectra · nucleosynthesis · cometary chemistry · quizzes.

### 3.7 Study tools

| Function          | Description                                          |
|-------------------|------------------------------------------------------|
| Spaced repetition | SM-2 algorithm on cards generated from the contents  |
| Quizzes           | By subject, with score and history                   |
| Study path        | A guided sequence of topics                          |
| Notes             | Global and per section                               |
| Pomodoro          | Study timer                                          |
| File Manager      | Personal archive of photos, documents and HTML pages |

### 3.8 The "Spectra" assistant

An AI agent with **35 tools** that can interrogate the application itself: compute descriptors, generate spectra, look up constants, balance reactions, open sections. It supports ten providers, and stays usable when one of them fails (§5.2).

## 4. Main flows

### 4.1 Analysing a molecule

1. Open the Spectroscopy centre
2. Enter the molecule:
  - SMILES → used directly
  - name or formula → resolution via PubChem (needs network)
  - drawing in the editor → SMILES generated
3. The application shows: 2D structure, functional groups, descriptors, predicted spectra (IR, NMR, MS, UV-Vis), drug-likeness
4. The 3D structure loads if there is a network (conformer from PubChem)

**With no network:** point 3 stays complete for SMILES entered directly; resolution by name and the 3D structure are unavailable, and the application says so instead of hanging.

### 4.2 Asking the assistant

1. Tap the "Spectra" button
2. On first use: choose a provider and enter an API key, or connect a proxy (and then no key is needed)
3. Type the question
4. The assistant answers, using its tools when needed: every tool used is shown, not hidden

**If the provider does not answer:** the assistant notes it, demotes it in the selection for 24 hours and moves to another configured provider. The  **Test** button measures in a few seconds which providers the device actually reaches, **without needing any key**.

### 4.3 Studying with spaced repetition

1. Open the study section
2. Generate a deck from the contents
3. For each card, rate how well it was remembered
4. The SM-2 algorithm computes when to show it again
5. On reopening, only the cards that are due appear

### 4.4 Archiving your own material

1. File Manager → local password
2. Upload photos, documents or HTML pages
3. The material stays ON THE DEVICE (IndexedDB), it is not transmitted

## 5. Business rules

### 5.1 Scientific data

| Rule                                                                                                                  | Rationale                                                                        |
|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| A molecular structure enters the database only if the recomputed weight agrees with the literature one ( $\pm 0.6$ u) | A plausible but wrong datum is indistinguishable from a correct one for the user |
| If a structure does not pass verification, the entry stays <b>without a structure</b>                                 | A wrong datum replaced by another wrong one is no progress                       |
| Every absence must be <b>recorded with its reason</b>                                                                 | An unjustified exception becomes a shortcut                                      |
| Spectra are deterministic                                                                                             | Two spectra of the same molecule must be comparable                              |
| An ambiguous physical constant is not resolved at random                                                              | A wrong value given with confidence is worse than a request for clarification    |

### 5.2 AI assistant

| Rule                                                                                   | Behaviour                                                  |
|----------------------------------------------------------------------------------------|------------------------------------------------------------|
| A provider that does not answer is remembered for <b>24 hours</b>                      | Then it is retried: today's failure is not a sentence      |
| A retired model is learned from the error message                                      | Hand-coded lists grow stale                                |
| An imposed wait longer than <b>10 seconds</b> switches provider, if a ready one exists | Beyond that threshold the user believes the app has frozen |
| One turn at a time                                                                     | Two overlapping submissions corrupt the history            |

| Rule                                        | Behaviour                                                       |
|---------------------------------------------|-----------------------------------------------------------------|
| Paid providers are not offered as free ones | Without active billing one hits the limit at the first question |

### 5.3 User data

| Rule                                        | Behaviour                                                                                                                             |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| At most 30 conversations, 100 messages each | Beyond that, <code>localStorage</code> fills up and <b>every</b> subsequent save fails                                                |
| Images up to 20 MB                          | Beyond that, explicit refusal; the others proceed                                                                                     |
| Selective deletion by group                 | Wiping everything to free space is an avoidable loss                                                                                  |
| A failed write <b>must</b> be declared      | "  Saved" when it is not true is worse than an error |

## 6. Current limitations

Declared, not hidden. The detail is in [09-Release-Conformance-Statement.md](#) §4.

| # | Limitation                                            | Practical consequence                                                                                                                                                                                                                                                                                                                                                |
|---|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>2 entries with no structure to show</b> (it was 6) | Ivermectin is a mixture of B1a/B1b homologues and Coartem a combination of two active ingredients: a single SMILES notation does not represent them, and for these two the 2D/3D and predicted spectra are missing. Digoxin, vincristine and tacrolimus (topical and systemic) now have their structure, taken from ChEMBL and verified against the molecular weight |
| 2 | <b>Predicted, not measured, spectra</b>               | Useful for recognising functional groups; not for formal identification                                                                                                                                                                                                                                                                                              |
| 3 | <b>No synchronisation</b>                             | Data do not pass from one device to another, and there are no copies on a server                                                                                                                                                                                                                                                                                     |
| 4 | <b>File Manager: deterrent, not security</b>          | On a static site, whoever reads the source bypasses any page-side control. The source carries only the SHA-256 digest of the password, never the password                                                                                                                                                                                                            |
| 5 | <b>Functions that need a network</b>                  | IUPAC name, CAS, GHS, 3D conformers, AI assistant                                                                                                                                                                                                                                                                                                                    |
| 6 | <b>Accessibility automated, not complete</b>          | Contrast, accessible names, labels, heading hierarchy across 13 pages and all 88 sections: <b>0 defects</b> , including text over gradients (judged against the worst stop). What stays outside is text inside SVGs and text over a real background image, counted on every run, and everything requiring human judgement                                            |
| 7 | <b>Chromatography: no retention prediction</b>        | The section computes resolution, efficiency and indices <b>given</b> $k$ , $\alpha$ and $N$ ; it does not predict $k$ from a structure, which would require experimental parameters of the stationary phase                                                                                                                                                          |
| 8 | <b>Partial code coverage</b>                          | Measured: <b>49.89 %</b> of statements over the widest path a bench walks. It is not whole-battery coverage, and it is statements, not branches; see <a href="#">08-Traceability-Matrix.md</a> §8                                                                                                                                                                    |

| # | Limitation                           | Practical consequence                                 |
|---|--------------------------------------|-------------------------------------------------------|
| 9 | <b>Verification on Chromium only</b> | Firefox and WebKit are tested by hand, not by a bench |

## 7. Possible developments

Not commitments: directions consistent with the architecture.

| Area          | Proposal                                                                        | Why                                                                                                                                       |
|---------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Data          | The 2 remaining entries have no structure to show                               | It is not a gap to close: ivermectin is a mixture, Coartem a combination. At most one could show the <b>components</b> , declared as such |
| Spectra       | Comparison with experimental databases (SDBS/NIST) inside the app               | It would make the distance between prediction and measurement visible                                                                     |
| Accessibility | Add <code>axe-core</code> alongside, for the rules the bench does not implement | ARIA roles, tab order and focus management stay outside the current measurement                                                           |
| Export        | Full export of the user's data                                                  | The only possible remedy for the absence of backups                                                                                       |
| Assistant     | A preconfigured proxy for those who do not want to publish one                  | It would remove the last barrier to AI access                                                                                             |

\_Document updated to version `bsi-v176` . \_